

Temporal Knowledge Graph

Formal definition

This formalization defines the graph's composition and meaning, then defines reasoning and matching in terms of that meaning.

A **temporal knowledge graph** is a tuple

$$\mathcal{K} = (\Sigma, \mathcal{O}, \mathcal{A}, \mathbb{T}, \Gamma)$$

with the following components:

Component	Definition
Σ	A signature containing individual names, class names, object-property names and data-property names, including vocabulary for processes and temporal entities.
$\mathcal{O}$	Ontology axioms specifying the meaning of the vocabulary and constraining its interpretation.
$\mathcal{A}$	A finite set of identified assertions concerning class membership, relations, data values and process occurrence, including temporally qualified assertions.
$\mathbb{T}$	A temporal structure specifying time positions, ordering, intervals and permitted temporal operations.
Γ	Explicit temporal constraints on the positions and intervals referenced by the assertions.

Temporal structure. For the basic model, let time positions be rational coordinates on a declared reference axis:

$$\mathbb{T} = (\mathbb{Q}, <, \delta), \quad \delta(s, t) = t - s.$$

Coordinates use a fixed unit, such as seconds. Proper intervals are half-open:

$$\text{Int}(\mathbb{T}) = \{[s, e) \mid s, e \in \mathbb{Q}, s < e\}.$$

A comparison between values from different clocks requires an explicit mapping into a common reference frame. An unknown mapping does not license a temporal comparison. Uncertain positions are represented by variables constrained by Γ , rather than assigned arbitrary timestamps.

Interpretation. A model $\mathcal{M}$ has an entity domain Δ , a data-value domain Δ_D , and an interpretation $\mathcal{I}_t$ at each time t . Individual names denote the same entities across time. Classes and properties are interpreted as:

$$\begin{aligned} C^{\mathcal{I}_t} &\subseteq \Delta, \\ R^{\mathcal{I}_t} &\subseteq \Delta \times \Delta, \\ D^{\mathcal{I}_t} &\subseteq \Delta \times \Delta_D. \end{aligned}$$

The vocabulary declares which classes and properties are **rigid**, with time-invariant extensions, and which may vary with time. For example, the classification of a particular process as a specimen collection may be rigid, while a patient's occupancy of a bed varies with time.

Ontology axioms concerning rigid vocabulary constrain its invariant interpretation. Axioms with a pointwise temporal interpretation must hold in every $\mathcal{I}_t$. Any additional connections between domain properties and temporal predicates require explicit bridge axioms.

Assertion interpretation. A temporally qualified relation assertion

$$R(a, b)@[s, e)$$

is satisfied when

$$\forall t \in [s, e) : (a^{\mathcal{M}}, b^{\mathcal{M}}) \in R^{\mathcal{I}_t}.$$

Class-membership and data-property assertions are interpreted analogously. An assertion at a single instant is evaluated in the interpretation at that instant.

This qualification asserts that the relation holds throughout the stated interval. It does **not** assert that the relation fails outside that interval. An unknown interval is represented by constrained variables; it is distinct from a declaration that a predicate is rigid.

Process occurrence is represented separately. Each process p has an occurrence interval $\omega^{\mathcal{M}}(p)$, and

$$\text{Occurs}(p, I) \iff \omega^{\mathcal{M}}(p) = I.$$

Thus, an assertion that a process occurred during I is a statement about that process's temporal extent. It does not expire when I ends. A recorded timestamp alone need not determine the complete occurrence interval.

Temporal predicates. Temporal relations and functions have explicitly specified interpretations. For intervals $I = [s_I, e_I)$ and $J = [s_J, e_J)$, examples include:

$$\begin{aligned} \text{Before}(I, J) &\iff e_I < s_J, \\ \text{Meets}(I, J) &\iff e_I = s_J, \\ \text{During}(I, J) &\iff s_J < s_I \wedge e_I < e_J, \\ \text{Overlaps}_{\text{Allen}}(I, J) &\iff s_I < s_J < e_I < e_J, \\ \text{Duration}(I) &= e_I - s_I, \\ \text{Gap}(I, J) &= s_J - e_I. \end{aligned}$$

Here, **During** denotes strict containment, Allen overlap is directional, and **Gap** is a signed difference. These meanings are part of the language specification.

Entailment and reasoning. Let $\text{Mod}(\mathcal{K})$ be the interpretations satisfying $\mathcal{O}$, $\mathcal{A}$ and Γ under the temporal structure. Then

$$\mathcal{K} \models \varphi \iff \forall \mathcal{M} \in \text{Mod}(\mathcal{K}) : \mathcal{M} \models \varphi.$$

Ontology reasoning establishes entailed class memberships, relation assertions and other ontological consequences. Temporal reasoning establishes entailed temporal relations or bounds and determines whether temporal constraints are jointly satisfiable. Both use the same notion of logical consequence.

Operational query answering requires a nonempty model set. If no model satisfies the selected knowledge, the system reports inconsistency rather than treating arbitrary statements as valid answers.

Trajectory patterns and matches. A trajectory pattern is a formula

$$\Psi(\mathbf{x}) = \Psi_{\text{ontology}}(\mathbf{x}) \wedge \Psi_{\text{time}}(\mathbf{x}),$$

where the first component contains class descriptions and relation conditions, and the second contains temporal conditions. Time-varying domain predicates must carry explicit temporal arguments or scope.

A binding μ assigns named entities or data values to the pattern's variables. It is a **certain match** when

$$\mathcal{K} \models \Psi[\mu].$$

It is a **possible match** when at least one model satisfies $\Psi[\mu]$. A possible match that is not certain remains unresolved under the available knowledge. The matching binding supplies the alignment between the query variables and the represented entities.

Process-Role-Object (PRO) pattern. Participation in a particular capacity is represented by linking a process to a role with `sulo:hasParticipant`, and that role to its bearer with `sulo:isFeatureOf`. The domain class `PatientRole` is a subclass of `sulo:Role`; `DrugAdministration` and `BloodSpecimenCollection` are subclasses of `sulo:Process`. In the formula below, `hasParticipant` and `isFeatureOf` denote these SULO properties; the clinical classes belong to the application ontology.

For an antibiotic administration a and subsequent blood specimen collection b , let r_a and r_b denote patient roles borne by the same individual p :

$$\begin{aligned} \Psi(p, a, b, r_a, r_b) = & \text{AntibioticAdministration}(a) \\ & \wedge \text{BloodSpecimenCollection}(b) \\ & \wedge \text{PatientRole}(r_a) \\ & \wedge \text{hasParticipant}(a, r_a) \\ & \wedge \text{isFeatureOf}(r_a, p) \\ & \wedge \text{PatientRole}(r_b) \\ & \wedge \text{hasParticipant}(b, r_b) \\ & \wedge \text{isFeatureOf}(r_b, p) \\ & \wedge \text{Before}(\omega(a), \omega(b)) \\ & \wedge \text{Gap}(\omega(a), \omega(b)) \leq 48 \text{ hours.} \end{aligned}$$

The two role bindings may differ: the query joins the processes through the common bearer p , without requiring the same role individual in both processes. A match retains the process, role and bearer bindings in its explanation. These assertions identify the capacity in which the individual participates in each process.

Ontology entailment through PRO. The drug-administration example also uses the PRO pattern. Let the application class `AdministeredDrugRole` be a subclass of `sulo:Role`. An antibiotic administration can be defined by a participating administered-drug role whose bearer is an antibiotic:

$$\begin{aligned} \text{AntibioticAdministration} \equiv & \text{DrugAdministration} \sqcap \\ & \exists \text{hasParticipant} . (\text{AdministeredDrugRole} \\ & \sqcap \exists \text{isFeatureOf} . \text{Antibiotic}). \end{aligned}$$

An administration linked to an instance of `AdministeredDrugRole`, borne by an instance of `Antibiotic`, can therefore be inferred to belong to `AntibioticAdministration`. Temporal reasoning and evaluation establish the ordering and gap conditions in the trajectory query.

The SULO participation chain is

$$\text{hasParticipant} \circ \text{isFeatureOf} \sqsubseteq \text{hasParticipant} .$$

Thus, $\text{hasParticipant}(a, r_a)$ and $\text{isFeatureOf}(r_a, p)$ entail $\text{hasParticipant}(a, p)$. This consequence establishes participation of the bearer, but does not preserve the identity or type of the intervening role. The trajectory query consequently retains the explicit `PatientRole` conditions and both links. Participation alone does not establish participation as a patient. The chain follows the SULO ontology, where the inverse of `hasFeature` is `isFeatureOf`.

A returned patient must have named process and role bindings that jointly satisfy the pattern. No additional domain-specific object properties are introduced by these examples.

Provenance and revision history annotate the identified assertions. For this project, a declared snapshot, source-information cutoff and revision policy select the knowledge to which these semantics are applied. Controlled relaxation modifies the query and assigns a cost to that modification; it does not change what the original knowledge entails.

Scope and sources. This is a proposed temporal extension, drawing on the interpretation and entailment framework of [OWL 2 Direct Semantics](#) and the temporal vocabulary described in [OWL-Time](#). It is not supplied automatically by timestamp annotations in OWL. The implementation must declare a supported language fragment; decidability, soundness and completeness must be established for that fragment.